

Practice Problems List-13 (Pointers)

1.
 - a) Write a C++ program to print the address of an integer variable without storing it to a pointer variable.
 - b) Write a C++ program to store the address of an integer variable to a pointer variable. Print the address from the pointer variable. Also print the value of the variable from its pointer.
 - c) Write a C++ program to change the value of a variable using its pointer.
 - d) Implement a function `printAddressAndValue()` which takes the reference (address) of an integer variable and prints the reference as well as the value stored to it.
 - e) Implement a function `isEven()` which takes the reference of an integer variable and returns true if the integer is an even number, else returns false.
 - f) Implement a function `convertToFahrenheit()` which takes the reference of a double variable signifying a temperature in Celsius and returns the corresponding Fahrenheit value of the temperature.
 - f) Implement a function `isLeapYear()` which takes the reference of an integer variable and returns true if the year is a leap year, else returns false.
2.
 - a) Write a C++ program to add two numbers using pointers.
 - b) Implement a function `addTwoNumbers()` that takes the references of two integer variables as input and returns the sum of the two integers.
 - c) Implement a function `maxNumber()` that takes the references of two integer variables as input and returns the sum of the two integers.
 - d) Implement a function `calculateAsYouWish()` that takes references to two numbers as well as the reference to a character. If the character is a '+', the function returns the sum of the two integers. If it is a '-' (minus), it returns the difference, and so on. The function must return the result as a floating-point number.
 - e) Implement a function `myPow()` that accepts the references to two integer variables and returns the value of x^y using pointers.
3.
 - a) Write a C++ program to swap two numbers using pointers.
 - b) Implement a function `mySwap()` that takes the references of two integer variables as input and swaps their values.
4.
 - a) Declare an integer variable. Store its reference to an integer pointer. Now, declare another pointer variable that stores the address of the integer pointer. Finally print the value of the integer from the second pointer.

- b) Implement a function `extractValue()` that takes the address of an integer pointer variable and returns the value of the integer.
5. a) Write a C++ program to print the elements of an array using a pointer.
b) Implement a function `printArray()` that receives the references to an array and prints the elements of it.
c) Implement a function `doubleUp()` that takes the reference of an integer array and doubles (multiplies by two) each of the elements of the array.
6. a) Write a C++ program to copy one array to another using pointers.
b) Implement a function `copyArray()` that receives the references to two arrays and copies the elements of one to the other.
7. Write a C++ program to swap two arrays using pointers.
8. Write a C++ program to reverse an array using pointers.
9. Write a C++ program to search an element in an array using pointers.
10. Write a C++ program to access the element at index `[0][0]` in a two dimensional array using pointers.
11. a) Write a C++ program to print each of the characters in a string using pointers.
b) Implement a function `printString()` that receives the references to a string and prints each of the characters in a new line.
12. Write a C++ program to swap two strings using pointers.